

**ECON 101—Introduction to Microeconomics
Fall 2019**

Final Exam

November 27, 2019

Last Name:

First Name:

NU ID number:

Instructions:

- No phones or other electronic devices except simple calculators are allowed.
- No books, notebooks or notes are allowed.
- **Cheating is not tolerated. Any instance of academic dishonesty implies an automatic grade of 0 in the final exam and the application of the penalties described in the Student Code of Conduct.**
- The exam has a duration of **150 minutes (2.5 hours)**
- Answer all questions.
- Record all your answers on this page.

Part I

Please record your answers here. Each question is worth 3 points.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Part II

Please record your answers here. Each question is worth 4 points.

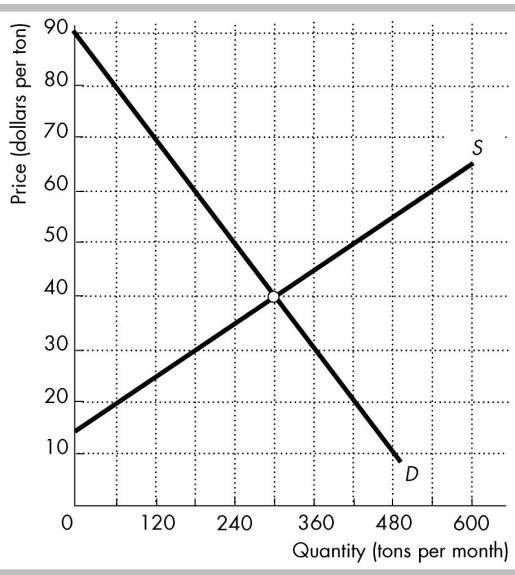
1	2	3	4	5	6	7	8	9	10

MCQ 1 (60)	MCQ 2 (40)	Total (100)
For instructor's use only. Do not write anything in here.		

Part I

Please record all questions in the cover sheet. Each question is worth 3 points for a total of 60 points.

1. Perfect price discrimination
- (a) cannot result in profit maximization.
 - (b) turns all the producer surplus into consumer surplus.
 - (c) creates a deadweight loss.
 - (d) turns all the consumer surplus into economic profit.



The figure shows the unregulated market for a pesticide, where S is the supply curve and D is the demand curve. The demand curve is the same as the MSB curve. When factories produce the pesticide, they also create waste, which they dump into a lake on the outskirts of the town. The marginal external cost of the dumped waste is equal to the marginal private cost of producing the pesticide.

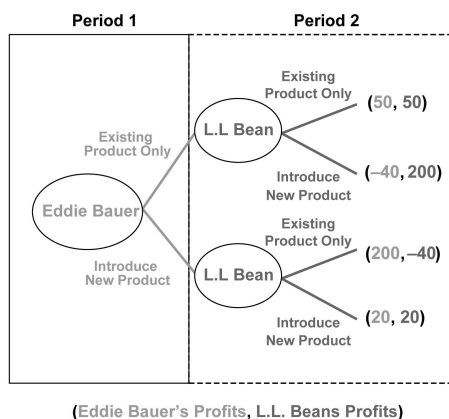
2. In the figure above, what is the efficient quantity of pesticide to produce?
- (a) 360 tons per month
 - (b) 300 tons per month
 - (c) 180 tons per month
 - (d) 240 tons per month

Quantity (lamps)	Marginal benefit for Mike	Marginal benefit for Joe	Marginal benefit for Sam
0			
1	4	8	5
2	3	6	4
3	2	4	3
4	1	2	2
5	0	0	0

3. Mike, Joe, and Sam are roommates. They need lamps for their common living room. Lamps in their living room are a public good for these roommates. The marginal benefit that each individual receives from varying numbers of lamps in the living room is given in the table above. If the marginal cost of a lamp is 9, what is the efficient number of lamps for the roommates to purchase?
- (a) 4
 - (b) 2
 - (c) 3
 - (d) 1
4. Four people each have a different willingness to pay for one unit of a good: George will pay \$15, Glen will pay \$12, Tom will pay \$10, and Peter will pay \$8. If price is equal to \$9 per unit then the quantity demanded in the market will be _____ and the consumer surplus for this unit will be _____.
- (a) 3; \$37
 - (b) 3; \$10
 - (c) 3; \$36
 - (d) 4; \$8
5. When the price of milk is \$3 per bottle, Steve consumes 20 bottles of milk. When the price increases to \$6, Steve’s consumption falls to 15 bottles. Therefore, Steve’s arc elasticity of demand for milk is
- (a) -0.50
 - (b) -0.75
 - (c) -0.25
 - (d) -0.43

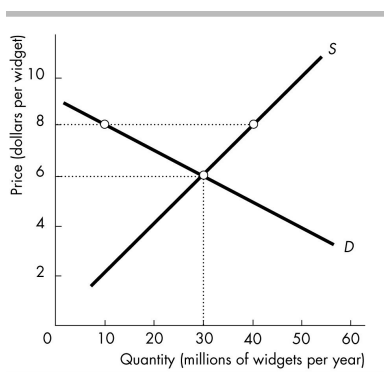
6. Suppose two outdoor apparel firms, L.L. Bean and Eddie Bauer, are contemplating introducing the same new product to their existing product lines. For this problem, assume that Eddie Bauer “moves” (makes this decision) first, and then L.L. Bean makes its decision. Each firm can choose to either introduce the new product or continue with the existing product. Eddie Bauer moves first, and there is “full information”, meaning that each firm knows the potential payoffs (profits associated with each action). In the extensive form of this sequential game, shown in the figure below, profits (shown in parentheses in the graph) are annual profits in thousands of dollars.

Sequential Game: Firms Deciding to Introduce New Product



Given the information above, which of the following statements is true?

- (a) L.L. Bean should choose to maintain the existing product only if Eddie Bauer chooses to introduce the new product.
 - (b) Eddie Bauer should always choose to maintain the existing product only.
 - (c) L.L. Bean should choose to introduce the new product if Eddie Bauer chooses to introduce the new product.
 - (d) L.L. Bean should choose to maintain the existing product only if Eddie Bauer chooses to maintain the existing product.
7. If the demand for gem-quality diamonds decreases AND the supply of gem-quality diamonds decreases, what can be said about equilibrium price and quantity in this market?
- (a) Equilibrium quantity will rise, but the change in equilibrium price is uncertain.
 - (b) Equilibrium price and quantity will both fall.
 - (c) Equilibrium quantity will fall, but the change in equilibrium price is uncertain.
 - (d) Equilibrium price and quantity will both rise.

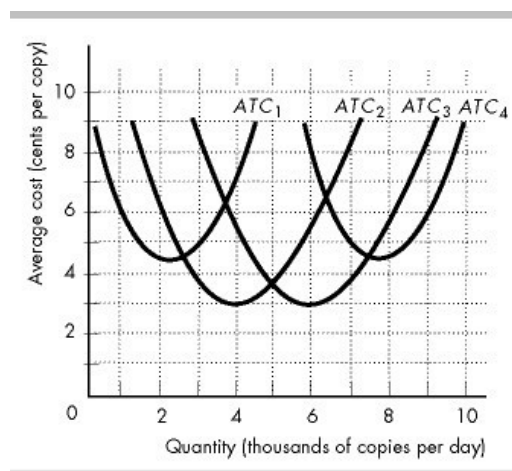


8. In the figure above, if a tax of \$2 per widget is imposed on sellers, then the price of widgets paid by the buyer will be
- (a) \$6.
 - (b) less than \$6.
 - (c) between \$8 and \$6.
 - (d) more than or equal to \$8.

Quantity (millions of vaccinations)	Perceived marginal benefit (dollars)	Marginal social benefit (dollars)
1	12	18
2	11	16
3	10	14
4	9	12
5	8	10
6	7	8

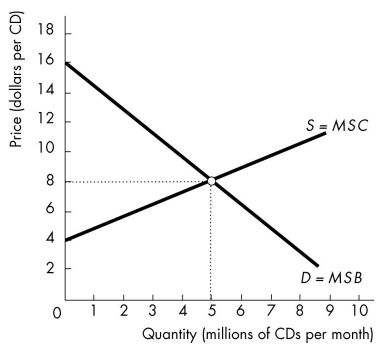
9. The table above gives information on the perceived marginal (private) benefit and marginal social benefit associated with vaccination against varicella (chicken pox). If the marginal cost of a varicella vaccination is \$10, then the efficient quantity of vaccinations is

- (a) 6 million vaccinations. (c) 3 million vaccinations.
(b) 5 million vaccinations. (d) 4 million vaccinations.



10. Dustin's copy shop can use four alternative plants. The figure above shows the average total cost curves for Plant 1 (ATC_1), Plant 2 (ATC_2), Plant 3 (ATC_3), and Plant 4 (ATC_4). What is Dustin's long-run average cost if the output is 3,000 copies per day?

- (a) 5.0 cents per copy
(b) 3.7 cents per copy
(c) 8.5 cents per copy
(d) More information is needed to determine the long-run average cost.



11. In the figure above, when the price of a CD is \$8.00, total producer surplus from all the CDs will be

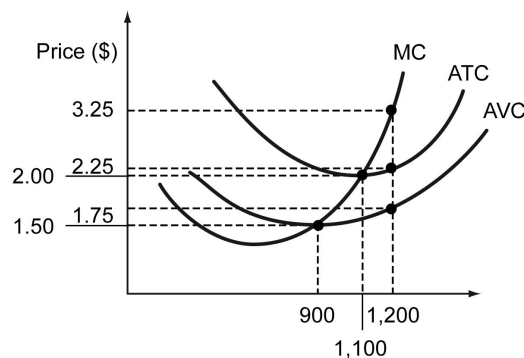
- (a) greater than at \$10.00 per CD. (c) zero.
(b) \$10 million. (d) \$20 million.

Scenario: Maylin is shopping at the sporting goods store for socks and running shoes. She has determined the following total benefit derived from each quantity of socks and running shoe

Pairs of Socks	Total Benefit, Socks	Marginal Benefit, Socks	Price of Socks	MB/Price	Running Shorts	Total Benefit, Running Shorts	Marginal Benefit, Running Shorts	Price of Running Shorts	MB/Price
0	0	NA	\$10	NA	0	0	NA	\$10	NA
1	100		\$10		1	200		\$10	
2	190		\$10		2	380		\$10	
3	270		\$10		3	540		\$10	
4	340		\$10		4	680		\$10	
5	400		\$10		5	800		\$10	
6	450		\$10		6	900		\$10	
7	490		\$10		7	980		\$10	
8	520		\$10		8	1040		\$10	
9	540		\$10		9	1080		\$10	
10	550		\$10		20	1100		\$10	

12. If Maylin has \$150 of income to spend on socks and running shorts, socks cost \$10 per pair and running shorts are \$10 each, what consumption bundle maximizes Maylin's total benefit?
- (a) 6 pairs of socks and 9 running shorts
 - (b) 1 pair of socks and 6 running shorts
 - (c) 3 pairs of socks and 7 running shorts
 - (d) 5 pairs of socks and 8 running shorts
13. A firm might be tempted to cheat on a collusive price-fixing agreement by setting a _____ price and producing _____ than agreed upon.
- (a) higher; less
 - (b) higher; more
 - (c) lower; less
 - (d) lower; more

The figure below shows the cost curves of a firm in a competitive market. The firm always makes the choice to maximize its profit.



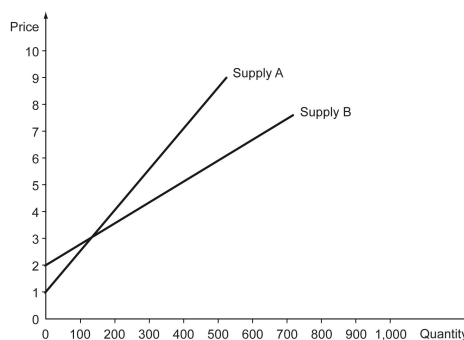
14. Refer to the figure above. The short-run supply curve of the firm is _____.
- (a) the upward-sloping portion of the marginal cost curve only
 - (b) vertical at 0 units between the price \$0 and \$1.50, then it follow the marginal cost curve
 - (c) vertical at 0 units between the price \$0 and \$2.25, then it follows the marginal cost curve
 - (d) the downward-sloping portion of the average variable cost until it crosses the marginal cost curve, then it follows the marginal cost curve

Price (dollars per repair)	Quantity demanded (repairs per week)	Total cost (dollars)
100	0	400
90	10	800
80	20	1400
70	30	2200
60	40	3200

15. Dee's TV Repair is the only TV repair shop in a small town. Dee is a single-price monopolist. Based on the demand and cost information in the table above, what is the amount of economic profit made or loss incurred at the quantity of TV repairs at which profits are maximized or losses minimized?

(a) -\$400 (b) \$200 (c) \$800 (d) -\$100

16. The supply curve of good A, and the supply curve of good B are both depicted in the following graph. Which statement is true?



(a) Supply of good A is more elastic than the supply of good B.
 (b) Supply of good B is perfectly inelastic.
 (c) Supply of good B is more elastic than the supply of good A.
 (d) Supply of good A is unit elastic.

17. The rate at which Sam is willing to give up a gallon of gasoline to get one more pound of coffee, and remain on the same indifference curve is called his

(a) opportunity cost of gasoline. (c) opportunity cost of coffee.
 (b) marginal rate of substitution. (d) personal price of coffee.

18. When the price of margarine is \$2 per pound, 10 pounds of butter are demanded. When the price of margarine increases to \$6 per pound, 30 pounds of butter are demanded. What is the cross-price elasticity between the two goods?

(a) -1 (b) 5 (c) 2 (d) 1

	Price elasticity of supply	Price elasticity of demand
Hamburgers	1.2	1.8
French fries	1.7	1.4
Pizza	2.0	1.2
Ice cream	1.5	1.6

19. You are in the business of producing and selling hamburgers, French fries, pizza, and ice cream. The mayor plans to impose a tax on one of these products. Based on the elasticities in the above table, as a profit-minded business person who seeks to avoid taxes whenever possible, which good would you *most* prefer to have taxed?

(a) pizza (b) hamburgers (c) ice cream (d) French fries

Scenario: The payoff matrix given below shows the payoffs to two firms in millions of U.S. dollars for choosing two alternative strategies. The first number listed in each cell is the payoff to the row player, and the second number listed is the payoff to the column player.

		Firm B	
		Strategy X	Strategy Y
Firm A	Strategy X	2.4, 2.6	2.4, 0
	Strategy Y	5.2, 3.0	4, 0

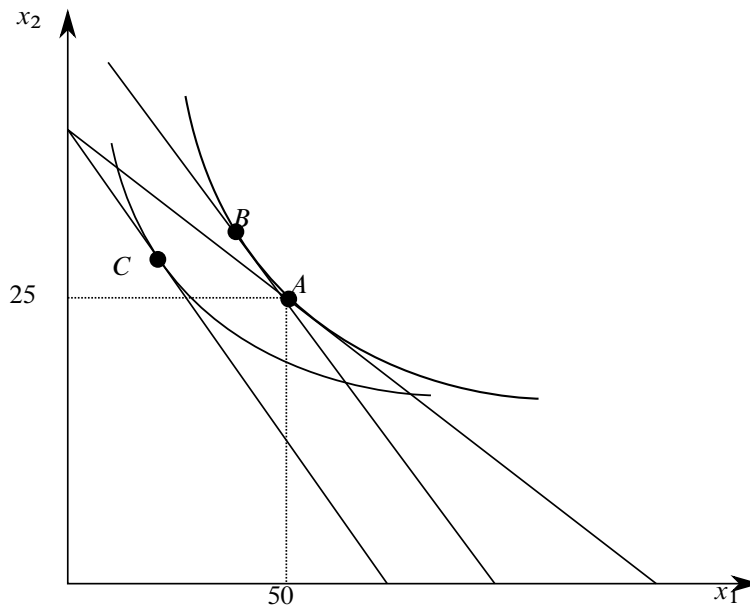
20. Refer to the scenario above. Which of the following is true in this case?

- (a) The dominant strategy equilibrium of this game is not Pareto efficient.
- (b) The dominant strategy equilibrium of this game is also its Nash equilibrium.
- (c) This game has multiple Nash equilibria.
- (d) This game does not have a Nash equilibrium.

Part II

Please record all questions in the cover sheet. Each question is worth 4 points for a total of 40 points.

Consider the following scenario as depicted by the following figure.



This consumer can consume goods x_1 and x_2 . The consumer has an income of \$100 dollars. The price of x_1 is \$1 dollar and the price of x_2 is \$2. Each consumer initially chooses the initial consumption bundle $x_1 = 50, x_2 = 25$ which is denoted by point A. Suppose that there is an increase in the price of x_1 to \$4. After this price increase, the optimal consumption bundle of the consumer is denoted by point C. Suppose the government gives the consumer a monetary transfer of \$80 so that she can achieve her original indifference curve before the price change. Also, assume that an economic researcher in the bureau of taxation gives you the following information:

- The arc own price elasticity for good 1 of the consumer is $|\epsilon_{x_1}^{p_1}| = 1.11$.
- The arc income elasticity with respect to good 1 of the consumer is $\epsilon_{x_1}^I = 1.5$

1. The substitution effect is:

- (a) -30 (b) -50 (c) -25 (d) -15

2. The income effect is:

- (a) -20 (b) -10 (c) -30 (d) -15

3. The arc cross-price elasticity of good x_2 with respect to price p_1 is:

- (a) -0.30 (b) 0.15 (c) 1 (d) 0.2

In some states of the United States of America, legislation has changed, giving the now the same treatment to marijuana and alcohol. The consumption, production and exchange of marijuana with recreational purposes is now legal.

Suppose that initially the demand and supply of marijuana were the following

$$Q_D = 1000 - 2P$$

$$Q_P = -100 + 2P$$

With the new legislation the production costs of marijuana have decreased and the demand has also increased so that the equations of demand and supply after the approval of the new legislation are:

$$Q_D = 1200 - 2P$$

$$Q_S = 2P$$

With the previous information answer the following three questions:

4. If the objective of the state governments is that the same quantity of marijuana is consumed before and after the legislation comes into force, the per unit tax on marijuana needs to be _____, a _____% of the tax will need to be paid by producers, and _____% of the tax will need to be paid by consumers.
 - (a) \$150; 50%; 50%.
 - (b) \$150; 75%; 25%.
 - (c) \$300; 50%; 50%.
 - (d) \$300; 75%; 25%.
5. After the drug legalization, which of the following quantities maximizes government revenues? The percentage of the tax that will be paid by producers is _____% and the the percentage paid by consumers is _____%.
 - (a) 225; 50%; 50%.
 - (b) 225; 75%; 25%.
 - (c) 300; 50%; 50%.
 - (d) 300; 75%; 25%.
6. Which of the following options minimizes the dead-weight loss associated with the imposition of a tax on marijuana?
 - A tax of \$200 per unit.
 - A tax of \$300 per unit.
 - A tax of \$150 per unit.
 - A tax of \$100 per unit.

Suppose that there are two firms, each with a total cost function $TC(q) = 10q$ and facing a demand function of $P = 130 - 2Q$, where Q is total output. The marginal cost of the firm is of course $MC(q) = 10$.

7. Suppose that the firms are operating as a cartel, dividing evenly profits and the aggregate level of production. The monopoly price is _____, each firm produces _____, and each firm earns profits of _____.
- (a) \$10;30;\$0
 - (b) \$70;15;\$900
 - (c) \$60;17.5;\$876
 - (d) \$30;25;\$500
8. Now suppose that the firms have two possible strategies: to produce half of the cartel aggregate quantity or to produce 20 units. If both firms produce 20 units, the market price is _____ and each firm earns _____ in profits.
- (a) \$30;\$500
 - (b) \$70, \$900
 - (c) \$50; \$800
 - (d) \$60; \$876
9. If one firm is producing half the cartel aggregate quantity and the other is producing 20 units, the market price is _____, the firm producing half the cartel quantity has profits of _____, and the firm producing 20 units has profits of _____.
- (a) \$876;\$900
 - (b) \$500; \$600
 - (c) \$200; \$1200
 - (d) \$750; \$1000
10. The Nash equilibrium of the game is _____.
- (a) For each firm to produce 20 units.
 - (b) For each firm to produce 15 units.
 - (c) For each firm to produce 30 units.
 - (d) For each firm to produce 10 units.

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